

RerankEverything: A Reranking Interface for Exploring Search Results

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ABSTRACT

This paper proposes a system called “RerankEverything”, which enables users to rerank search results in any search service, such as a Web search engine, an e-commerce site, a hotel reservation site, and so on. This system helps users explore diverse search results. In conventional search services, interactions between users and systems are quite limited and complicated. By using RerankEverything, users can interactively explore search results in accordance with their interests by reranking search results from various viewpoints. Experimental results show that our system potentially help users search more proactively. When using our system, users were more likely to click search results that were initially low ranked. Users also browsed through more diverse search results by reranking search results after giving various types of feedback with our system.

Categories and Subject Descriptors

H.5.2 [Information Interfaces and Presentation]: User Interfaces – *Graphical user interfaces, Interaction styles.*

General Terms

Design, Human Factors.

Keywords

Search user interfaces, information retrieval, reranking interaction.

1. INTRODUCTION

People access the World Wide Web for various purposes, such as acquiring knowledge, purchasing something, making travel plans, reserving hotels, or just watching interesting videos. To find the desired information, people use many kinds of search services such as Web search engines, e-commerce sites, hotel reservation sites, video sharing sites, and so on. When using a search service, users usually issue a query to it. The service then selects contents from its index, ranks them in accordance with its own criteria, and finally displays them as search results. These search services have become important information resources for many people.

However, it is not always easy for users to obtain satisfactory information from search results because of the following problems:

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- It is difficult for users to come up with appropriate queries that clearly describe their information needs. Since users' information needs are often not clearly defined, their queries are often too short and ambiguous. Therefore, it is difficult for search services to estimate users' information needs from only the received queries and rank results in ways that will satisfy all users.
- Some services provide advanced search functions such as filtering or sorting, so that users can refine search results in accordance with their needs. However, such functions are provided only by the *service-side*: a user cannot filter or sort search results from a certain viewpoint if a service does not provide such a function. Moreover, advanced search functions are not easy for most users to use because these interfaces are often complicated. In addition, the interfaces differ depending on the services.

Consequently, interactions between users and services are limited and complicated. This fundamental problem limits users to browsing search results passively. When the user's information needs are not well defined, it is especially important for the user to iteratively search and browse information [2]. If users were able to flexibly refine search results on the basis of their own criteria, they could obtain more useful content in accordance with their interests. Our goal is to realize such a *user-side* interactive search system.

In this paper, we propose a system called *RerankEverything*, which enables users to rerank search results in any search service. In the system, users can intuitively rerank results by simply selecting an interesting term, or a numeric value of a search result. Our system provides users not only the reranking interface, but also a tag cloud to encourage users to explore search results from various viewpoints, and a simple interface to specify an html element that contains a search result to recognize structures of the search results page.

2. SYSTEM ARCHITECTURE

We implemented our system (<http://rerank.jp/rerankeverything.xpi>) as an extension to the Mozilla Firefox Web browser. Our system mainly consists of three components.

2.1 Reranking Interface

In our system, users can give both term-based feedback and numerical feedback by using the same interaction style, simply selecting an interesting part of a search result (Figure 1). When a user selects a certain term and presses the promote/demote button, the system then reranks the results by promoting/demoting results that contain the selected term. Similarly, a user can sort search results according to a selected numerical attribute.

Figure 2 illustrates how the user reranks search results on a search results page of Google Scholar, a search engine specializing in



Figure 2. Reranking publication results with both term-based and numerical feedback in publication search. Users can give both term-based and numerical feedback to system and rerank search results by simply selecting area of interest and clicking appropriate button.

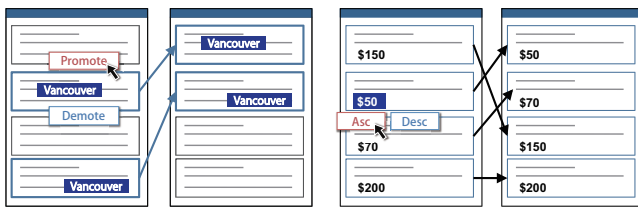


Figure 1. Reranking search results using direct interaction.

academic publications. First, the user issues the query “pseudo relevance feedback”. Then, the user starts browsing search results and takes an interested in papers authored by Wei-Ying Ma. The user selects “WY Ma” and presses the promote button to promote search results containing “WY Ma”. If the user wants to browse papers in descending order of the citation counts, the user selects “Cited by 181” (in this case, the user can select just “181” to sort results) in the search results, and the system then displays four buttons with which to rerank them. After the user clicks the descending button, the system sorts the search results in accordance with the number of citations, as shown on the right-hand image in Figure 2.

In this way, users can interactively rerank search results in accordance with their interests. Google Scholar does not provide search functions to sort results in accordance with the citation counts, although it does display these counts in the search results. Our system thus enables users to sort using this attribute even when the service does not provide such a sorting function.

2.2 TermCloud

To encourage users to explore search results from various viewpoints, our system displays *TermCloud*, which are generated from terms that frequently appear in search results. Users can give term-based feedback and rerank search results by simply clicking a term in a *TermCloud*. Figure 3 shows an example of a *TermCloud* of Web search results for the query “pork green pepper recipes.” In that case, the system displays some terms that are related to cooking like “garlic”, “onion”, “Chinese”, “easy” and so on. Because a *TermCloud* might bring interesting and various terms in search results to the attention of users, we expect users will be more likely to find valuable and diverse search results.

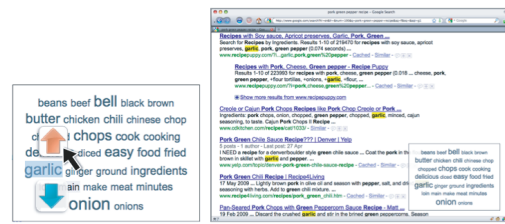


Figure 3. Example *TermCloud* for Web search results of query “pork green pepper recipes”.

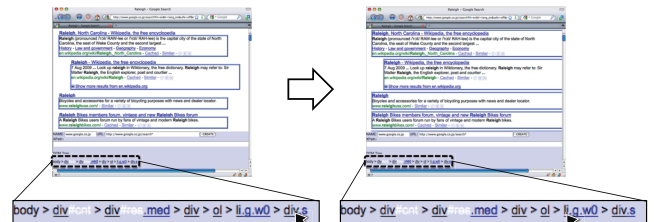


Figure 4. Wrapper generation interface.

2.3 Wrapper Generation Interface

The number of existing search services is huge and the structures of search results vary according to the services. To detect a structure of search results in an HTML source of a target page, we need to prepare wrappers for each service. Our system provides an interface to generate wrappers based on simple user interactions. The interface also works interactively inside a Web browser and requires no special knowledge of DOM, JavaScript, XPath, etc.

Our idea is to visualize nodes that the system currently recognizes as search results while a user moves the mouse cursor. Figure 4 illustrates how a user would generate a wrapper that works on the Google Web search engine. As the user moves the mouse cursor, the system displays visual feedback by forms of rectangle indicating which nodes the system recognizes as search results.

3. USER STUDY

To investigate how our system affects users’ search behavior, we conducted two user studies. The first evaluated the usefulness of our system compared with the conventional search service in the specific search task. The second investigated how users use our system in their daily browsing.

3.1 Task-based Study

To evaluate how our reranking interaction supports users seeking information, we conducted a user study. We think our reranking interaction become important when the user's information needs are not well defined and the user needs to browse search results from diverse viewpoints. Therefore, we prepared such search tasks and compared the effectiveness of performing the tasks between our system and conventional search services.

We recruited eight students from our university as participants: four graduate students, and four undergraduate students. All were male. The subjects did our experiment individually. In the study, we first asked the subjects to watch a three-minute-long instruction video (<http://rerank.jp/everything/demo.mp4>) that explained how to rerank search results. After they watched the video, we asked the subjects to visit a Web search engine and spend five minutes learning how to use our system. Then we asked the subjects to perform the following tasks:

- You are seeking recipes for tomorrow's dinner. Use the Web search engine with our system and find as many interesting recipes that contain pork and green pepper as possible in five minutes.
- After that, use the Web search engine without using our system and find as many interesting recipes that contain beef and onion as possible in five minutes.

During the task, when the subjects found a search result that contained information about an interesting recipe, they were asked to double-click it, which changed its background color. In the experiment, if necessary, the subjects were allowed to perform all kinds of interactions that the Web search engine provided, such as following links from search results, modifying their queries, using advanced search options, clicking suggested queries, and so on, regardless of whether they were using our system or not. The wrapper for the Web search engine was prepared in our system in advance so the subjects did not need to create any wrappers.

To prevent the order effect, we divided eight subjects into four groups and changed the order of queries ("pork and green pepper" and "beef and onion") and interfaces for searching for the information ("using our system" and "without using our system").

3.1.1 Results

From the study, we can see that the numbers of selected search results highly depended on the subjects. For example, one user chose many results (67 results) using our system and another user chose much fewer results (4 results) using our system. Average numbers of search results that subjects selected when using our system were not significantly different from those without using our system ($t_{14}=0.37$, $p>.05$). However, we can see that 6 out of 8 subjects can find more results when they used our system. These results show that our system could support users in finding more interesting results than conventional search engines alone. In the conventional search engines, the ways for users to state their interests are limited to query modification or advanced search options. On the other hand, in addition to these functions, users can state their interests by selecting or clicking terms in the search results and TermCloud by using our system. Thus, users could easily collect search results that contain interesting terms like ingredients and adjectives by performing reranking operations.

When the subjects used only the Web search engine, the average number of query modifications during the task was 0.625 (five query modifications within the whole subjects.) Four modifications were the modifications that the subjective adds

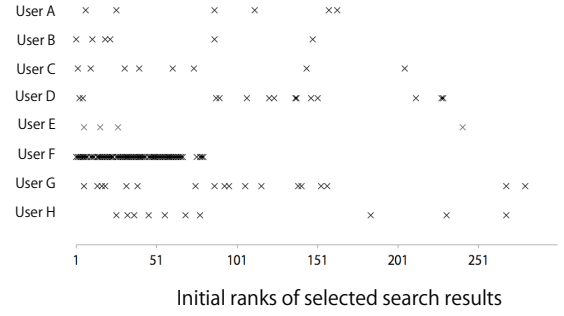


Figure 5. Distributions of initial ranks of selected search results (using our system).

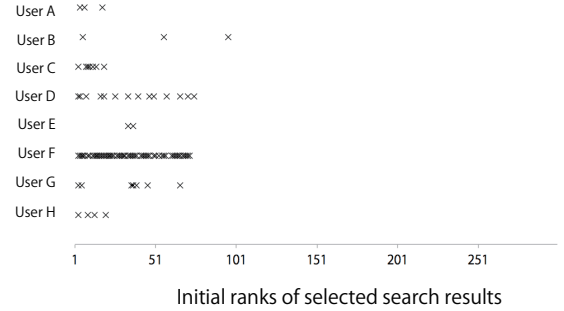


Figure 6. Distributions of initial ranks of selected search results (without using our system).

some keywords to the previous queries. One modification was the modification that the subject replaces some keywords in the previous query to another keyword. This result indicates that users tend to *narrow down* the information space when they use the Web search engine alone. On the other hand, the average number of reranking operations was 6.25 when they used our system. These results indicate that our system helps users to *explore* search results from diverse viewpoints. Figures 5 and 6 respectively show the distributions of initial ranks of search results that the subjects judged to be interesting with and without using our system. The initial rank of a search result means the rank of the search result before the subjects performed any reranking operation. Comparing the results in Figures 5 and 6, we found that the subjects were more likely to find originally low-ranked search results by using our system. These low-ranked search results are usually hard to find by using only conventional search engines. This result indicates that when using our system, users are more likely to browse different search results by reranking search results, even if they issue the same query. Although the tasks in this study were limited and the number of subjects was small, the results of this study indicate that our system can potentially enable users to search more proactively. In the future, we plan to conduct this kind of study with more diverse search tasks and more diverse subjects.

3.2 Longitudinal use of Our System

Second, to evaluate how our system would affect users' daily search behavior, we conducted a user study. The purpose of this study was to investigate how users interacted with various search results in our system.

Seven subjects participated in this study. Two subjects had participated in the experiments discussed in the previous user study. Of the remaining five subjects, four were graduate students and one was a faculty member of the university. All subjects were

active Web researchers and reported using Firefox as their default Web browser.

In the study, we first asked the subjects to watch the instruction video described above. After they watched the video, we asked them to install our prototype system on their Firefox browser for the user study. The system already contained several wrappers for major search services such as Google, Yahoo!, Bing, Amazon, and so on. The subjects were allowed to use our system for seven days. In the study, we did not impose any specific search tasks on the subjects. We asked the subjects to freely use our system since we wanted to investigate how our system would affect users' daily search behavior.

3.2.1 Results

During the test period, we collected 408 reranking operations from the subjects. The distribution of the types of reranking operation and the areas where users performed reranking operations are shown in Table 1. From the results, we can observe that about 70% of the reranking operations were promotion operations. One subject performed promotion operations eight times on one search results page. On the other hand, demotion operations were rarely used. As much literature has explained [1], it was difficult for users to give negative feedback. The total queries issued to search engines were 452, and 17.7% queries were reranked. We think this percentage is not small since most queries that were not reranked were navigational query and users did not need to rerank search results.

Half of the term-based feedback was done in the TermCloud area. This indicates the TermCloud was effective in suggesting terms for reranking search results. When the subjects used Web search engines, 62% (94/151) of term-feedbacks were performed in TermCloud. On the other hand, when the subjects used other search services, such as publication search service and e-commerce sites, only 32% (51/150) of term-feedbacks were performed in TermCloud. Web search results consisted of more unstructured text such as titles and snippets, so the subjects might have had difficulty in finding interesting terms from the search results. Therefore, for Web searching, the subjects selected terms in TermCloud rather than in search results. Moreover, such unstructured text contains various terms, so TermClouds generated from Web search results could display unexpected and interesting terms to the subjects.

Numerical feedback was used for various services such as e-commerce sites, publication sites, video sharing sites, and so on. For example, the number of views and the uploaded dates were used on video sharing sites to sort video search results. The citation count and the publication year were often used on publication search services. For Web searching, some subjects tried to sort search results using numerical feedback. For example, one subject issued the query "iPod" and sorted search results by selecting "64GB" in a search result. Thus the search results containing "160GB" were reranked higher. In this way, our dynamic attribute extraction method allowed the subjects to utilize Web search results as ad hoc structured resources. Overall, however, the subjects rarely used numerical feedback in Web searching (only 22 times). We think this was because sorting Web search results was an unfamiliar task for the subjects and it was not easy for them to find appropriate numerical attributes from unstructured Web search results. Moreover, our extraction method is simple, so it often fails to extract correct attributes from Web search engines. To address these problems, we plan to develop some kinds of TermCloud-like suggestion systems that will suggest numerical attributes for sorting search results and a more

Table 1. Distributions of reranking operations.

Reranked Area	Reranking operation types				Total
	Term-based		Numerical		
	Pro mote	De mote	Asc	Desc	
Search results	138	18	15	92	263
TermCloud	131	14	-	-	145
Total	269	32	15	92	408

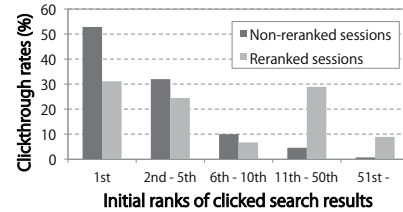


Figure 7. Comparison of clickthrough rates.

sophisticated extraction method using surrounding text of the selected text.

Finally, to investigate what kind of search results the subjects clicked, we measured the clickthrough rates in our system. To evaluate the effectiveness of reranking, we divided all sessions into two groups: non-reranked sessions and reranked sessions. To detect a session unit, when a subject visited a search result page, our system automatically assigned a session ID to the page. Non-reranked sessions represent sessions in which the subjects did not perform any reranking. Reranked sessions represent sessions in which the subjects performed at least one reranking. Figure 7 shows the clickthrough rates obtained from these two groups. We found the results to be similar to those from the previous study.

4. CONCLUSION

We have developed and evaluated a system called "RerankEverything", which provides unified and simple interaction techniques to enable users to rerank search results. Our experimental results showed that our system enables users to browse and explore search results more proactively. From system logs of the user study, we found that users reranked search results from various viewpoints and that they were more likely to click initially low-ranked search results when using our system. These kinds of search activities can potentially help users find more diverse information.

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